

CS 15/L - First Laboratory Exam - SET A

Scenario:

You are tasked to develop a simple **Smart Home Mobile Application** using React Native.

The application will allow the user to view smart-home devices and navigate to a separate screen containing information about a selected device.

Students are expected to demonstrate proper use of:

- Flexbox for layout
- StyleSheet or styled-components for styling
- React Navigation for screen navigation

Minimum Requirement:

Application runs without errors.

1. APPLICATION SETUP

Clone the repository given to you by the teacher.

Rename React Native application to:

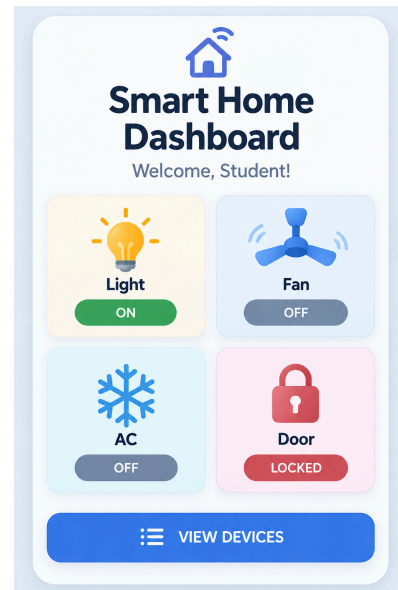
YourLastName-SmartHomeApp

The application must contain the following screens:

HomeScreen
DevicesScreen
DeviceDetailsScreen

2. HOMESCREEN

Create a Home Screen similar to the following layout:



Requirements

- Flexbox - The four device cards must be displayed in **two columns**.
- Layout

The following must be visually organized:

- Header
- Welcome message
- Device cards
- Button

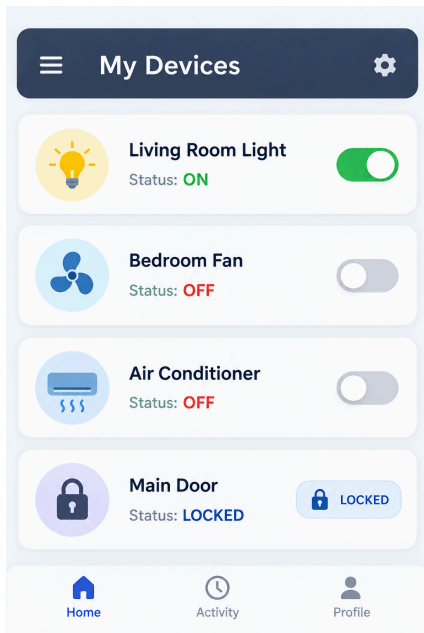
- Styling

The screen should have:

- Appropriate font sizes
- Appropriate spacing
- Card borders/background
- Readable text
- Proper alignment

3. DEVICES SCREEN

Create a screen displaying all available smart-home devices.

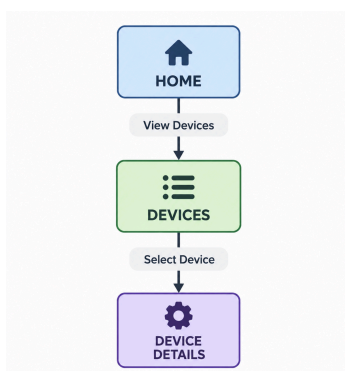


Requirements

1. Display four devices
2. Proper layout using Flexbox
3. Proper styling using StyleSheet/styled-components
4. Each device is selectable

4. REACT NAVIGATION

Configure navigation between the three screens.



Requirements:

- Stack Navigator

Configure the three screens in a stack navigator.

Home

Devices

DeviceDetails

- Home → Devices

The **VIEW DEVICES** button must navigate to the Devices screen.

- Devices → Device Details

When the student selects a device, the application must navigate to **DeviceDetailsScreen**.

The selected device information must be passed through navigation.



- Back Navigation

The user must be able to return to the previous screen.

5. STYLING REQUIREMENTS

Stylesheet or styled-component

Criteria	Excellent	Satisfactory	Needs Improvement	Poor/Not Submitted	Points
1. Flexbox Layout	Uses Flexbox correctly and efficiently; layout is organized, responsive, and closely follows requirements	Uses Flexbox correctly with minor layout issues	Uses some Flexbox properties but has noticeable layout problems	Little/no proper Flexbox implementation	25
2. StyleSheet / styled-components	Styling is properly structured, consistent, readable, and produces a polished interface	Styling is generally correct with minor inconsistencies	Styling works but is poorly organized or has several inconsistencies	Styling is largely missing or incorrectly implemented	20
3. React Navigation Setup	Navigation is fully configured and all required screens work correctly	Navigation works with minor issues	Some screens/navigation work, but several issues exist	Navigation is missing or largely non-functional	20
4. Navigation Between Screens	All required navigation paths work correctly, including back navigation	Most navigation paths work correctly	Some navigation paths work	Navigation does not work properly	15
5. Passing Data Between Screens	Correctly passes and displays selected device information using navigation parameters	Passes data with minor errors	Attempts to pass data but implementation is incomplete	No working data transfer	10
6. Overall Application Functionality	Application runs smoothly without errors and all requirements are functional	Application runs with minor errors	Application runs but has several functional issues	Application cannot run or major features are missing	5
7. Code Organization	Code is clean, logical, readable, and appropriately separated into components/screens	Code is generally organized with minor issues	Code is difficult to follow or has unnecessary repetition	Code is poorly organized or incomplete	5

TOTAL

100